

Let's citrusize pangenome graphs.

BioHackathon series:

[DBCLS BioHackathon 2026](#)

Matsuyama, Japan, 2026

[mikan-pangenome](#)

Submitted: 18 Sep 2026

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First Author¹ and Last Author ²

1 First Affiliation 2 ELIXIR Europe 

Introduction

As part of the DBCLS BioHackathon 2026, we here report...

Meeting information

If you want to submit a preprint to BioHackrXiv, first check if your meeting is registered. You can find a list of meetings [here](#). If your meeting is missing, please contact your meeting organizers. The above list also provides information on the YAML fields with information about the meeting.

The following fields need to be given:

```
biohackathon_name: "DBCLS BioHackathon 2026"
biohackathon_url:  "https://2026.biohackathon.org/"
biohackathon_location: "Matsuyama, Japan, 2026"
group: YOUR-PROJECT-NAME-GOES-HERE
git_url: https://github.com/yourOrganization/your_report_repo
```

The [BioHackrXiv meeting pages](#) provide content to use for the first three fields. The `git_url` field must have the link to the GitHub repository with your preprint (draft).

Author information

Information about the authors is given in the [YAML](#) format at the top of this template. For authors you provide their names, their affiliations. That is the minimum, but as BioHackrXiv is moving to a situation where more metadata is shared, and used by, for example, EuropePMC, adding additional information is encouraged.

BioHackathons is about hacking together, and the minimal number of authors for reports is two. This makes a minimal example look like this:

```
authors:
  - name: First Author
    affiliation: 1
  - name: Last Author
    affiliation: 2
affiliations:
  - name: First Affiliation
    index: 1
  - name: ELIXIR Europe
    index: 2
```

Author identifiers

Ideally, authors provide their [ORCID](#) identifier. For affiliations, It is added with the `orcid` field. So, an author record would look like this:

```
authors:  
  - name: First Author  
    affiliation: 1  
    orcid: 0000-0000-0000-0000
```

Research Organization Registry identifiers

Matching the author identifier, the affiliations can be further specified with the [Research Organization Registry](#) (ROR) identifier. For example, this is the affiliation identifier can be added with the `ror:` field:

```
affiliations:  
  - name: ELIXIR Europe  
    ror: 044rwnt51  
    index: 2
```

Contributor Role Taxonomy

A last feature since is minimal support for the Contributor Role Taxonomy (CRediT). You can specify the role of authors in writing the report with the `role:` field. However, the authors are responsible for selection the right terms from [CRediT](#). An example looks like this:

```
authors:  
  - name: First Author  
    affiliation: 1  
    orcid: 0000-0000-0000-0000  
    role: Conceptualization, Writing - review & editing
```

A full examples

A full example then has this structure:

```
authors:  
  - name: First Author  
    affiliation: 1  
    role: Writing - original draft  
  - name: Last Author  
    orcid: 0000-0000-0000-0000  
    affiliation: 2  
    role: Conceptualization, Writing - review & editing  
affiliations:  
  - name: First Affiliation  
    index: 1  
  - name: ELIXIR Europe  
    ror: 044rwnt51  
    index: 2
```

Formatting

This document use Markdown and you can look at [this tutorial](#).

Subsection level 2

Please keep sections to a maximum of only two levels.



Tables

Tables can be added in the following way, though alternatives are possible:

Table: Note that table caption is automatically numbered and should be given before the table itself.

```
| Header 1 | Header 2 |
| ----- | ----- |
| item 1  | item 2  |
| item 3  | item 4  |
```

This gives:

Table 1: Note that table caption is automatically numbered and should be given before the table itself.

Header 1	Header 2
item 1	item 2
item 3	item 4

Figures

A figure is added with:

```
![Caption for BioHackrXiv logo figure](./biohackrxiv.png)
```

This gives:



Figure 1: Caption for BioHackrXiv logo figure

Figures can be scaled by adding the width or height to the Markdown like this:

```
![Caption for BioHackrXiv logo figure](./biohackrxiv.png){ width=50px }
```

You can add cross references to figures by adding a LaTeX `\label{figureCode}` to the label of the Markdown figure and then use `\ref{figureCode}` to cite it:

```
![Caption for BioHackrXiv logo figure \label{figureCode}](./biohackrxiv.png){ width=50px }
```

This way, we can cite Figure 1.

Other main section on your manuscript level 1

Lists can be added with:

1. Item 1
2. Item 2

Citation Typing Ontology annotation

You can use [CiTO](#) annotations, as explained in [this BioHackathon Europe 2021 write up](#) and [this CiTO Pilot](#). Using this template, you can cite an article and indicate *why* you cite that article, for instance DisGeNET-RDF (Queralt-Rosinach et al., [2016](#)).

The syntax in Markdown is as follows: a single intention annotation looks like `[@usesMethod In:Krewinkel2017]`; two or more intentions are separated with colons, like `[@extends:discusses:Nielsen2017Scholia]`. When you cite two different articles, you use this syntax: `[@citesAsDataSource:Ammar2022ETL; @citesAsDataSource:Arend2022BioHackEU22]`.

Possible CiTO typing annotation include:

- `citesAsDataSource`: when you point the reader to a source of data which may explain a claim
- `usesDataFrom`: when you reuse somehow (and elaborate on) the data in the cited entity
- `usesMethodIn`
- `citesAsAuthority`
- `citesAsEvidence`
- `citesAsPotentialSolution`
- `citesAsRecommendedReading`
- `citesAsRelated`
- `citesAsSourceDocument`
- `citesForInformation`
- `confirms`
- `documents`
- `providesDataFor`
- `obtainsSupportFrom`
- `discusses`
- `extends`
- `agreesWith`
- `disagreesWith`
- `updates`

There is a general `cites` intention, but this is already implied and should be left out.

Results

Pangenome Viewer

To enable browser-based exploration of the chromosome-specific GFA files generated in the preceding analyses, we developed a program that extracts shared nodes, non-reference sequences, path orientations, within-path positions, and local similarity measures and converts them into lightweight JSON files. Using the Satsuma mandarin CUNphKu haplotype (CUN#1) as the reference coordinate system, we developed the Unshu Haplotype Viewer, which allows six haplotypes from Satsuma mandarin, Kishu mandarin, and Kunenbo mandarin to be compared within a common coordinate system (Figure X).

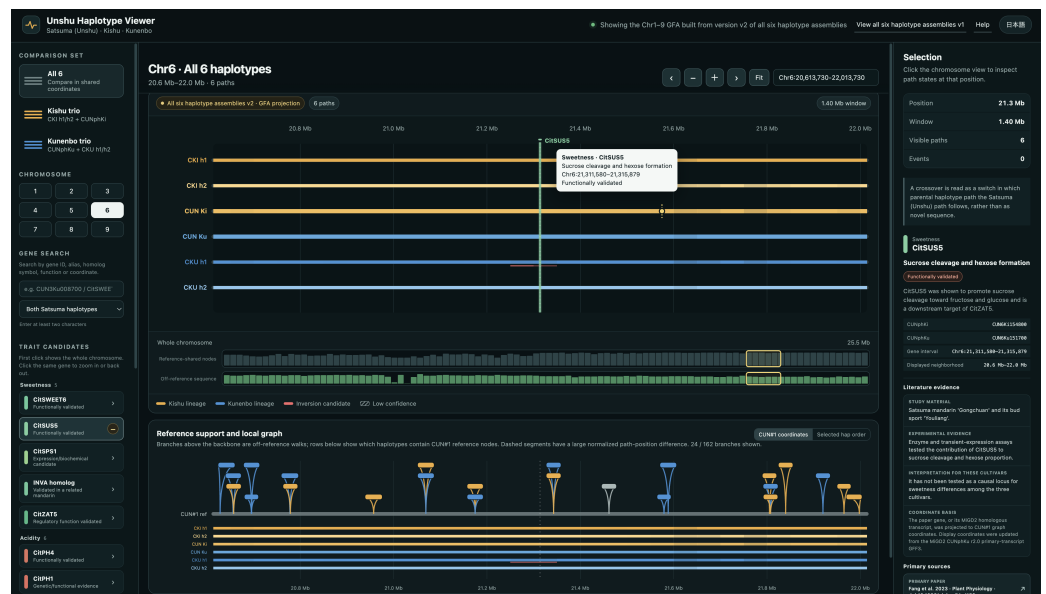


Figure X. The Unshu Haplotype Viewer. The viewer presents the six haplotype paths, reference-sequence support, non-reference branches, local similarity to the parental lineages, and candidate-gene search results within the same genomic interval.

The viewer provides both a combined display of all six haplotypes and two three-haplotype views that compare each Satsuma haplotype with either the Kishu or Kunenbo lineage. It supports chromosome-wide overviews, interval zooming, horizontal panning, reference-node coverage, candidate inverted regions, the amount of non-reference sequence, and local graph visualization. In the local graph, sequences following the reference path are distinguished from non-reference branches, and each branch is coloured according to the haplotypes that traverse it. Shared-node similarity between each Satsuma haplotype and the four parental haplotypes is also displayed along the chromosomes, allowing users to examine whether a region is more similar to the Kishu or Kunenbo lineage. These views facilitate visual exploration of candidate switches in parental origin, structural variation, and potential assembly or phasing inconsistencies.

To facilitate investigation of genotype–phenotype relationships, we registered literature-derived candidate genes associated with sweetness, acidity, bitterness, aroma, peel colour, fruit size, seedlessness, polyembryony, and bioactive compounds. We also constructed a search index containing 49,322 genes from the MiGD2 CUNphKi and CUNphKu v2 primary-transcript annotations. Users can navigate to any annotated gene by searching for a gene ID, transcript ID, homologous-gene symbol, functional description, or genomic coordinate. CUNphKu annotation coordinates are used directly, whereas CUNphKi coordinates are projected onto CUN#1 using shared GFA nodes. The viewer reports the projection method and shared-node coverage. Genes that cannot be placed on chromosomes 1–9 remain available in the search index.

A Help page describing the calculations and their interpretive limitations was included, together with a separate URL for accessing the previous assembly version. The viewer was published through GitHub Pages, and the source code, display data, and data-conversion programs were released in a public GitHub repository.

GitHub repository: <https://github.com/moriya-dbcls/unshu-haplotype-viewer>

Public viewer: <https://moriya-dbcls.github.io/unshu-haplotype-viewer/>

Shared-node similarity and path switching do not directly establish parental origin. They were therefore treated as exploratory measures for identifying candidate regions that require further validation through nucleotide-level alignment, structural-variant analysis, and HiFi read

support.

Database and visualization-tool development has traditionally emphasized generic designs intended for long-term reuse. In contrast, the principal components of this viewer were implemented within several hours through interaction with AI, demonstrating that interfaces tailored to a particular dataset and research question can now be developed rapidly. This does not mean that general-purpose design is no longer necessary. Standardization, maintainability, and reusability remain essential for infrastructure intended to support multiple studies and datasets over an extended period. For exploratory research with a restricted scope, however, rapidly developing a purpose-specific viewer may be more efficient. Future development should therefore distinguish between components that require a reusable, general-purpose foundation and those that are better implemented for a specific research question, according to the expected lifetime of the resource, its intended users, the diversity of the input data, and the goals of the analysis.

Discussion

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Acknowledgements

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References

Author contributions

First Author: Writing – original draft; Last Author: Conceptualization, Writing – review & editing.

References

Queralt-Rosinach, N., Pinero, J., Bravo, À., Sanz, F., & Furlong, L. I. (2016). DisGeNET-RDF: harnessing the innovative power of the Semantic Web to explore the genetic basis of diseases. *Bioinformatics*, 32(14), 2236–2238. <https://doi.org/10.1093/bioinformatics/btw214> [cito:citesAsAuthority].



Appendices

If you want the Appendix (-ces) to show up after the references, wrap them in after the header, like done in this Markdown file. Look at the [source](#) to see the exact structure.